RESEARCH PROJECT

Hardware-Software System for Sleep Phase Monitoring

« Bio-Headband 1.0 with "BioHeadband AI" Intelligent Bot »

Field: Bio-inspired and Neurotechnologies

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CRITERION 1. PROBLEM STATEMENT, OBJECTIVES, AND PROJECT GOALS

1.1. Problem Relevance

Problem: Sleep Deficit and Insufficient Accuracy of Existing Monitoring Methods

Contemporary adolescents between the ages of 13 and 18 are disproportionately affected by chronic sleep deprivation. According to the Sleep Foundation, approximately 73% of teenagers fail to obtain the clinically recommended quantity of sleep [1]. One of the principal contributing factors is the phenomenon of "digital displacement": blue-spectrum light emitted by smartphones and computer screens suppresses melatonin biosynthesis, thereby disrupting endogenous circadian rhythms. Without consistent monitoring of sleep architecture — particularly Rapid Eye Movement (REM) sleep — the neural restorative processes that occur during sleep are impaired. This degradation produces demonstrably negative consequences: diminished cognitive performance, elevated anxiety indices, and chronic fatigue syndrome [2].

Evidence-Based Scientific Rationale:

- Endocrine Impact: Blue-light exposure reduces melatonin secretion by up to 50%, increasing sleep-onset latency by 1.5 to 2 hours (Harvard Medical School) [3].
- Cognitive and Memory Impairment: NCBI research confirms that disruption of normal sleep architecture and reduction of REM phase duration suppresses new information consolidation capacity by approximately 40% [4].
- Technological Gap: The majority of commercially available consumer devices — smartwatches and fitness bands — employ indirect proxy methods such as heart rate (HR) monitoring or actigraphy. According to MIT Media Lab, such modalities are fundamentally incapable of accurately classifying sleep stages N1, N2, and REM, as they do not capture the direct electrophysiological activity of the brain [5].

1.2. Problem Description

Two categories of sleep analysis solutions currently exist in the consumer market, and neither adequately addresses the needs of the adolescent demographic:

Comparative Analysis of Existing Solutions:

Method / Device	Operating Principle	Primary Limitation	Key Shortcoming	Source
Mobile Apps (Sleep Cycle)	Acoustic analysis + smartphone accelerometer	High sensitivity to ambient noise; accuracy <50%	Cannot differentiate sleep stages	JCSM [6]
Smartwatches (Apple, Xiaomi)	Photoplethysmography (PPG) + accelerometer	No direct brain activity analysis	REM classification error 35–40%	Meltzer [8]

Method / Device	Operating Principle	Primary Limitation	Key Shortcoming	Source
Clinical PSG (in-hospital)	EEG + EOG + EMG full-channel array	"First-night effect"; patient discomfort	Inaccessible for daily home use	AASM: gold standard [7]
Bio-Headband 1.0 + BioHeadband AI	EEG + AI-based CSV analysis	Requires active noise & artifact filtering	Only viable consumer-grade neural pathway	MIT [5]

1.3. Justification for Project Development

The Bio-Headband 1.0 project was engineered to occupy the critical gap between low-cost, high-error consumer devices and prohibitively expensive clinical diagnostic infrastructure. The necessity for development is substantiated by three convergent factors:

- Consumer-grade sleep trackers exhibit classification errors of 35–40% [9]. The Bio-Headband's EEG-based methodology directly addresses limitations in measurement precision, electromagnetic artifact filtration [10], and intelligent sleep-stage-aware awakening algorithms [11].

1.4. Hardware Platform and Technical Specifications

The finalized prototype is built on the Arduino Nano microcontroller platform, programmed in C++ with full-stack digital signal processing (DSP) pipeline integration. The Zero-EMF design protocol ensures that all sleep data is recorded to CSV format via USB connection — entirely eliminating Bluetooth radio-frequency exposure during the recording session (compliant with ICNIRP safety guidelines).

1.5. Project Goal

To engineer an integrated hardware-software complex in C++ and Python for the automated acquisition, real-time processing, and intelligent AI-powered analysis of electroencephalographic (EEG) biosignals for automated sleep stage classification, delivered through the portable Bio-Headband 1.0 wearable device and the BioHeadband AI Telegram intelligent analytics bot.

1.6. Project Objectives

- Implement digital noise filtration for MPU6050 electromyographic and motion artifacts.
- Develop and deploy ATmega328P firmware for multi-modal biosignal acquisition.
- Engineer the BioHeadband AI Telegram bot (Python + OpenAI API) for automated CSV-based sleep analysis.
- Build an automated Serial Monitor-to-CSV data conversion pipeline.
- Validate the intelligent smart alarm algorithm under real-world sleep conditions.

1.7. Target Audience and Key Performance Indicators (KPIs)

Primary Target Audience:

- Adolescents and students aged 13–18 years (demographic most vulnerable to digital sleep displacement).

Potential Stakeholder Segments:

- Somnology clinics and sleep research centers: home pre-diagnostic screening instrument.
- Biohackers and cognitive performance professionals: precise sleep architecture analytics.
- Students and knowledge workers: rapid recovery optimization for irregular schedules.

Key Performance Indicators (KPIs):

Performance Indicator	Target Value	Verification Method
REM/NREM stage classification accuracy	≥ 80%	Comparative analysis against reference pulse oximeter benchmark
Device mass	< 300 g	Laboratory precision scale (±1 g)
Battery operational life	≥ 12 hours	Li-Po battery charge cycle testing
Smart wake-up alarm latency	< 30 minutes	Software-level event timestamp logging
Wearability comfort score (0–10 scale)	< 8 / 10	Subjective participant evaluation

CRITERION 2. REVIEW OF EXISTING SOLUTIONS AND METHODOLOGIES

2.1. Overview of Consumer Sleep Monitoring Devices

Category	Representative Products	Operating Principle	Limitations
Mobile Apps	Sleep Cycle, Pillow	Smartphone microphone + accelerometer	JCSM: 50% error rate; misclassifies movement as stage transitions [6]
Fitness Bands	Apple Watch, Xiaomi Band	PPG heart rate + accelerometer	NCBI: 40% REM phase error [8]
Smart Rings	Oura Ring, Withings	Core body temperature + heart rate analysis	Univ. Helsinki: confuses rest with true sleep [9]

Conclusion: While consumer-grade sleep tracking devices offer convenience, they consistently underperform in data quality. Their primary deficiency is reliance on indirect proxy biomarkers — heart rate, acoustic patterns, and movement. As MIT Media Lab conclusively demonstrates, the only viable pathway to accurate home sleep monitoring is direct registration of the brain's electrophysiological activity via EEG [5].

2.2. Analytical Methods Employed in Consumer Sleep Solutions

Method	Mechanism	Limitation	Source
Actigraphy	Accelerometer motion analysis; absence of motion inferred as sleep	Cannot distinguish rest from sleep; 40% error (Univ. Helsinki [9])	University of Helsinki, 2022
Photoplethysmography (PPG)	HRV algorithm analyzes inter-beat intervals	Sensitive to physiological stress and physical state variability	NCBI: REM accuracy ~60% [8]
Acoustic Analysis	Neural network isolates snoring and respiratory patterns	Ambient noise distorts signal; JCSM correlation <50% [6]	JCSM, 2021

Conclusion: All three modalities analyze downstream symptoms rather than the primary source. Sleep is fundamentally a neocortical phenomenon; heart rate and body movement are merely secondary physiological correlates. The next generation of consumer sleep technology must be anchored in direct brain rhythm analysis [5]. This principle forms the theoretical and engineering foundation of Bio-Headband 1.0.

2.3. Clinical Methodology — The Gold Standard: Polysomnography (PSG)

In clinical medicine, polysomnography (PSG) represents the only officially validated method for comprehensive sleep architecture assessment. It is administered exclusively in specialized sleep laboratories.

Parameter	Description	Barrier to Home Use
Sensor Count	15 to 22 sensors (EEG, EOG, EMG, ECG)	Severe patient discomfort; wiring restricts natural sleep posture
Accuracy	100% (clinical gold standard)	—
Cost	50,000–150,000 KZT per session; hospitalization required	Economically inaccessible for routine use
Analysis Method	Professional-grade software (Somnologica)	Requires trained somnologist for interpretation

Conclusion: PSG constitutes an ideal diagnostic tool for sleep disorder pathology, but is entirely unsuitable for routine daily monitoring. According to the American Academy of Sleep Medicine (AASM), the clinical environment itself induces the "first-night effect" [7], causing measurable divergence between laboratory sleep architecture and naturalistic home-environment sleep patterns.

2.4. Cross-Category Comparative Analysis

Criterion	Consumer Trackers	Clinical PSG	Bio-Headband 1.0
Sleep Stage Accuracy	65–70% (Low)	~95% (Gold Standard)	80–85% (Target)
Data Source	Movement, heart rate	19+ sensor array	EEG + HRV + Actigraphy
Analysis Method	Indirect inference	Direct (medical-grade)	Direct (neurometric)
EMF Safety	Bluetooth 2.4 GHz radiation	No radiation	Zero-EMF (ICNIRP-compliant)
Cost (KZT)	5,000–20,000	>2,000,000	~12,500
Wearability	High (wristband)	Low (hospitalization)	High (headband)

2.5. Identified Limitations of Existing Solutions

- Information Gap: Consumer-grade devices achieve only 65–70% accuracy because they analyze circulatory proxies rather than direct neurological data. End users receive hypnographic reports that carry a 40% probability of misclassification.

- "First-Night Effect": Clinical PSG requires 19+ simultaneous sensors, structurally disrupting natural sleep architecture and introducing systematic measurement bias [7].
- Electromagnetic Field Exposure: Wireless consumer devices emit Bluetooth radio-frequency radiation at 2.4 GHz in close proximity to the user's cerebral cortex across 8 hours of sleep — a quantifiable and avoidable health risk factor.
- Absence of Intelligent Artifact Filtering: The majority of commercial algorithms lack the capability to isolate neural signals from motion-induced noise artifacts without high-cost clinical-grade hardware.

2.6. Justification of the Proposed Solution

- Direct Neurometric Measurement: Bio-Headband 1.0 employs a single-channel EEG methodology to isolate alpha (8–12 Hz), theta (4–8 Hz), and delta (0.5–4 Hz) brain rhythms, achieving a target classification accuracy of 80–85%.
- Zero-EMF Protocol: Software-level radio module deactivation during sleep recording, implemented in compliance with ICNIRP electromagnetic safety standards.
- Economic Accessibility: Component cost of approximately 12,500 KZT — more than 160 times less expensive than clinical polysomnography.
- Sleep Inertia Mitigation Algorithm: The proprietary "smart wake-up window" algorithm ensures arousal during light sleep phases (N1/N2), preserving cognitive performance and post-sleep productivity [11].

CRITERION 3. METHODS, MATERIALS, AND RESOURCE PROVISIONING

3.1. Problem-Solving Methods and Software-Level Logic

Multimodal EEG + HRV Analysis with On-Device Microcontroller Processing:

- Digital Signal Processing (DSP) — Butterworth Filter Implementation: A software-implemented Butterworth bandpass filter was designed and validated to isolate delta-band rhythms (0.5–4 Hz) and theta-band rhythms (4–8 Hz) from the raw EEG signal stream, effectively suppressing electromyographic (EMG) motion artifacts that contaminate surface-electrode recordings.
- Time-Domain HRV Analysis: Computation of the RMSSD metric (Root Mean Square of Successive Differences between consecutive RR intervals) to quantify autonomic nervous system activity. RMSSD is a validated clinical marker for parasympathetic tone and sleep-stage-specific cardiac modulation.
- Smart Wake-Up Window Logic: A threshold-based algorithm continuously compares real-time HRV metrics against a personalized baseline computed from the first 30 minutes of each recording session. When HRV signature indicates a transition from deep (N3) to light (N1/N2) sleep, the smart alarm is activated within the defined comfort window.

3.2. Hardware and Software Selection Rationale

All component selections were driven by portability and cost-efficiency constraints:

- ATmega328P Microcontroller: Provides precision RR-interval timing resolution to 1 millisecond, enabling high-fidelity HRV computation.
- Development Environment: Arduino IDE with C++ language. Implementation of direct port manipulation (register-level I/O programming) in place of Arduino standard library calls reduced the signal processing loop execution time by 25%.
- Zero-EMF Protocol: Programmatic radio module deactivation embedded in the firmware — a biological safety measure that eliminates HF electromagnetic field exposure to the brain during the full recording session.

3.3. BioHeadband AI Software System

The BioHeadband AI intelligent Telegram bot constitutes the core analytical component of the system. Developed in Python, it provides fully automated CSV data processing and AI-powered user consultation.

3.3.1. System Architecture and Technology Stack

- python-telegram-bot Library: Primary framework for Telegram Bot API interaction — CSV file ingestion, response dispatch, and user interface state management.

- OpenAI API Integration: GPT model integration for intelligent sleep pattern analysis. A custom prompt engineering pipeline was designed specifically for sleep medicine data interpretation, training the model to recognize phase-specific biosignal patterns from structured CSV input.
- Pandas and NumPy: Scientific computing libraries for CSV preprocessing, descriptive statistical analysis, outlier detection, and feature extraction.

3.3.2. Bot Functional Capabilities

- CSV Sleep Log Analysis: User uploads a CSV export; the bot autonomously classifies sleep stages (REM / Deep / Light) and returns a personalized, evidence-based recommendation report.
- Analysis History: Persistent storage of all prior analyses enables longitudinal tracking of sleep quality trends over time.
- AI Consultation Chat: Interactive consultation interface where users submit sleep health inquiries. The AI responds with scientifically grounded, personalized guidance.
- "Authors & About" Module: Project information and development team documentation.

3.3.3. BioHeadband AI Processing Algorithm

1. CSV Ingestion: User transmits the CSV file to the bot via the Telegram interface.
2. Data Preprocessing: The pipeline validates file schema integrity, removes malformed or corrupted rows, and applies data normalization.
3. Statistical Feature Extraction: Computation of core biometric metrics: mean BPM, EEG signal variance index, movement event count.
4. OpenAI API Query: Processed feature vector is submitted to a custom-engineered system prompt; the language model performs pattern recognition and sleep stage classification.
5. Report Generation: The bot returns a structured clinical report including a hypnogram visualization and actionable sleep improvement recommendations.

3.4. Complete System Operating Cycle — Bio-Headband 1.0

6. Data Acquisition: The Bio-Headband 1.0 prototype — equipped with EEG electrodes, a pulse sensor, and a 3-axis accelerometer — continuously records multimodal biosignals throughout the sleep session.
7. Serial Data Transmission: The raw data stream is transmitted via Arduino Serial Monitor to the user's host computer over USB.
8. Automated CSV Conversion: The Python-based Serial-to-CSV converter processes the raw data stream into a time-stamped, structured CSV format.
9. Telegram Bot Upload: The user transmits the CSV file to BioHeadband AI via Telegram.
10. AI Analysis and Report: The bot analyzes the data and returns a complete report containing sleep stage classifications, sleep quality score, and personalized recommendations.

3.5. Resource Provisioning and Project Budget

The project was developed within a school laboratory environment. All computing resources were provided by the institution. Hardware components were procured with personal funds.

Resource / Material	Role in Project	Source	Cost (KZT)
Software (Arduino IDE, VS Code, Python)	Firmware development & debugging	Open Source	0
Hardware (Arduino Nano, sensors)	Data acquisition & algorithm execution	Personal funds	12,500
Libraries (Filters.h, I2Cdev, python-telegram-bot)	Signal processing foundation	Open Source	0
OpenAI API	AI-powered sleep data analysis	Personal API key	~1,000/month
		Total Project Cost:	**~13,500 KZT**

The minimal total cost confirms the high scalability and broad economic accessibility of the product concept.

CRITERION 4. PROJECT RESULTS AND EXPERIMENTAL VALIDATION

4.1. Prototype Deliverables and Functional Outcomes

The project has produced a fully functional Bio-Headband 1.0 hardware-software complex, including the BioHeadband AI Telegram bot.

Hardware Deliverable:

A wearable headband prototype (mass < 300 g; autonomous operation: 14 hours) integrating EEG surface electrodes, a pulse sensor, and a 3-axis MEMS accelerometer. The final production-equivalent version utilizes the Arduino Nano platform (Arduino Uno was used during early-stage development iterations).

Software Deliverable:

- Firmware (Hardware Layer): C++ firmware for the ATmega328P microcontroller implements multi-sensor data acquisition, Butterworth DSP filter processing, and real-time OLED display output.
- Data Pipeline (Software Layer): A Python conversion script processes the raw Arduino Serial Monitor output stream into structured, time-stamped CSV format. The BioHeadband AI Telegram bot ingests these CSV files, invokes the OpenAI API analysis engine, and returns comprehensive sleep reports with stage classifications and evidence-based recommendations.

Zero-EMF Implementation: Data recording conducted exclusively via USB connection — no Bluetooth transmission occurs during the sleep session.

4.2. Experimental Testing Protocol and Methodology

A structured test protocol was developed and executed to validate all declared project performance indicators:

#	Test Parameter	Test Conditions	Evaluation Method	Result
1	Prototype Functionality	Sensor attachment; firmware boot sequence	Visual inspection of OLED display output and Serial Monitor	PASS — All sensors initialized; real-time data stream confirmed
2	EEG Signal Analysis	Eyes closed; full muscle relaxation protocol	Frequency spectrum analysis of EEG output	PASS — Distinct alpha rhythm detected; peak at 10.2 Hz

#	Test Parameter	Test Conditions	Evaluation Method	Result
3	Serial Data Transfer	USB connection; continuous data recording session	CSV file integrity verification	PASS — Data stream intact; no packet loss
4	Battery Life	Continuous uninterrupted data recording	Li-Po charge monitoring over time	PASS — 14 hours of autonomous operation achieved
5	Device Weight	Final prototype assembly	Laboratory precision scale (±1g)	PASS — Below 300g threshold confirmed
6	BioHeadband AI Bot	Upload of standardized test CSV file	Report accuracy and content verification	PASS — Sleep stage classifications and recommendations verified correct

4.3. Evidence and Validation Documentation

Video Evidence:

- Bot demonstration — interface walkthrough, CSV upload, sleep analysis report generation
- Prototype worn by test subject — biosignal acquisition and Telegram upload workflow
- Full bot operational cycle — menu navigation → file upload → graph rendering → AI recommendation output
- AI Consultation Chat — sleep improvement advisory session
- Real-time device operation — sensor feed, OLED display, active algorithm execution
- Firmware development session — code authoring and display-driver bug resolution

Photographic and Screenshot Evidence:

- Sleep data graphs: EEG, pulse, and accelerometer traces rendered on host PC
- Device OLED display: real-time biometric output
- Sensor signal visualization: live monitoring interface

4.4. Alignment of Results with Project Goals

An EEG-based wearable instrument for automated sleep stage classification has been successfully developed, validated, and documented. The BioHeadband AI Telegram bot implements a complete end-to-end analytical pipeline from raw biosignal acquisition to personalized, AI-generated sleep health recommendations.

4.5. Practical Significance of Results

Bio-Headband 1.0 carries significant preventive health implications for adolescent sleep disorder intervention. At a total cost approximately 160 times lower than clinical alternatives, it democratizes access to neurometric sleep analysis for the demographic most susceptible to sleep disruption.

4.6. Current Limitations and Development Roadmap

The current version represents a validated functional prototype with the following acknowledged limitations:

- Large-Scale Clinical Validation Pending: Comprehensive accuracy validation requires multi-subject studies with direct comparison against medical-grade polysomnography.
- Algorithm Calibration: Filtering and classification algorithms require iterative refinement through expanded, demographically diverse user datasets.
- Ergonomic Enclosure Design: The prototype chassis requires industrial design refinement for long-duration wearability and potential commercial production.

Development Roadmap:

- Conduct full clinical trials with 10–20 participants to generate statistically significant validation datasets.
- Advance machine learning classification algorithms to achieve 85–90% sleep stage accuracy.
- Develop a native mobile application to simplify the device-to-bot data transfer workflow.
- Pursue medical device certification for clinical-use authorization.

CRITERION 5. INDEPENDENCE, TECHNICAL DEPTH, AND TEAM CONTRIBUTION

The Bio-Headband 1.0 project was executed by a two-person team. A clearly delineated division of engineering responsibilities enabled concurrent parallel development of hardware and software subsystems — a methodology that accelerated prototype delivery and enhanced overall system reliability.

5.1. Team Role Assignment and Contribution Matrix

Team Member	Role	Primary Contributions
Aisultan	Software Engineer	ATmega328P firmware architecture design (C++); interrupt-driven RR-interval measurement; Butterworth DSP filter implementation; HRV-based sleep stage classification algorithm; Serial-to-CSV Python conversion pipeline; BioHeadband AI Telegram bot (full-stack); OpenAI API integration and prompt engineering; end-to-end software testing.
Akzhol	Hardware Engineer	Sensor and controller component selection and procurement; electrical circuit schematic design; PCB soldering and hardware assembly; ergonomic headband chassis fabrication; sensor mount engineering; hardware validation and physical testing.

5.2. Author's Personal Contribution — Software Engineering Portfolio

The following technical deliverables were designed, implemented, tested, and validated independently by the author:

Video Materials in Russian:

- **Bot Demo:** Interface, CSV upload, and sleep analysis — [https://drive.google.com/file/d/1jeZ6DQW6DksuOpKPC650xm3bw6qlzG_1/view?usp=sharing]

- **Prototype on a Subject:** Data collection and transmission to the bot — [https://drive.google.com/file/d/19djzDqUz-G-mIiuFRlKcWOekvFmi1Qi/view?usp=sharing], [https://drive.google.com/file/d/19QFtAH8CvXQ8fffSmZRZzvUenal8oRqV/view?usp=sharing]

- **Full Bot Workflow:** Menu → Upload → Charts → AI Recommendations — [https://drive.google.com/file/d/1AJAkEJ0YGPo7Z4-xIFj54fCVRLi108A/view?usp=sharing]

- **AI Chat:** Sleep improvement consultations — [https://drive.google.com/file/d/19eQxKbexWaEH11bU_rO9eyGnzN84pPym/view?usp=sharing]

- **Real-time Device Operation:** Sensors, display, and algorithms — [https://drive.google.com/file/d/1daE7Jeewac2iDvH1SYsG_2i7l4jkFDjZ/view?usp=sharing]

- **Code Execution:** Writing and debugging process (fixing a display error) — [https://drive.google.com/file/d/14-6M9F4loINNaQ4Ymnjce15rUR4XcM7k/view?usp=sharing]

Photos/Screenshots in English:

- **Sleep Charts:** EEG, Heart Rate, and Accelerometer data on PC — [https://drive.google.com/file/d/1gmaD7YDsbUEoQ4tTgEMzhEfaaFWaLhmu/view?usp=sharing]

- **Device Display:** Real-time data visualization — [https://drive.google.com/file/d/1-QoH01JRdBOSQRCFqwZsXIxvYwQCOR5T/view?usp=sharing]

- **Sensor Signals:** Signal visualization in monitoring software — [https://drive.google.com/file/d/1HNk_KiPqT0GLVA0l2Wd3epyxyBV3A7X6/view?usp=sharing]

Interrupt-Driven RR-Interval Measurement

Designed and implemented a hardware interrupt service routine (ISR) on the ATmega328P for high-precision RR-interval capture, achieving sub-millisecond temporal resolution required for clinical-grade HRV computation.

Digital Signal Processing — Butterworth Filter Implementation

Independently designed, coded, and validated a software-implemented Butterworth IIR bandpass filter in C++. The filter attenuates electromyographic (EMG) artifacts and motion-induced noise from the raw EEG input stream, isolating neurologically relevant delta, theta, and alpha frequency bands with high selectivity.

HRV-Based Sleep Stage Classification Algorithm

Drawing on peer-reviewed HRV analysis literature, engineered a real-time sleep stage transition detection algorithm. The core logic: when the computed RMSSD metric drops below a dynamically

calculated personalized threshold (derived from the initial 30-minute baseline window), the firmware logs a confirmed transition into deep sleep (N3). This rule-based classifier operates autonomously in embedded real-time.

Automated Serial-to-CSV Data Conversion Pipeline

Developed a Python script that automatically ingests the raw ASCII data stream from Arduino Serial Monitor, applies timestamp injection, performs schema validation, and exports structured time-series CSV files for downstream AI analysis.

BioHeadband AI Telegram Bot — Full-Stack Development

Independently architected and implemented the complete BioHeadband AI system. Technical scope included:

- Full-stack Python application architecture using the python-telegram-bot framework.
- OpenAI API integration with a custom sleep-science-specific prompt engineering pipeline — the system prompt instructs the GPT model to recognize biosignal patterns characteristic of REM, N1, N2, and N3 sleep stages from structured CSV feature vectors.
- Pandas/NumPy data preprocessing module for CSV validation, outlier removal, and statistical feature extraction (mean BPM, EEG variance index, RMSSD, motion event frequency).
- Persistent analysis history module for longitudinal sleep quality tracking.
- Interactive AI consultation interface for evidence-based, personalized sleep health advisory.
- Complete user interface design, conversation state management, and end-to-end integration testing.

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